

B5 T7-SCOPE assignment: admissibility-bounded beyond-layer closure (operator promotion 2026-06-13)

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope

Enactment record for the operator route-3 promotion (2026-06-13): B5 $\rightarrow$ T7-SCOPE on the admissibility-bounded statement. **Scope:** the B5 claim tier and its scope qualifier. **Not claimed:** B5 unrestricted / global / arbitrary- Q ; any change to T-030 (which stays OPEN).

2. The promotion and the mandatory scope qualifier

B5 canonical statement (A), now T7-SCOPE. For every admissible competitor $Q \in \mathcal{C}_{\text{adm}}$ (real antipodal, packing-bounded $|Q| \leq n_{\text{pack}}$, coherence-resolved pairwise geodesic $\geq \theta_{\text{min}}$), given A1-KERNEL-CONV, the beyond-layer off-diagonal correction is bounded below the layer margin. Tier: **T7-SCOPE**_{admissibility-bounded}; substantive conditional hypotheses: NONE; definitional scope: $\mathcal{C}_{\text{adm}} + \text{A1-KERNEL-CONV}$.

Mandatory scope qualifier (anti-overclaim). B5 does NOT claim the unrestricted arbitrary- Q DR-2 bound $E_+(Q) \leq_\epsilon N^{2+\epsilon}$; it claims ONLY the admissibility-bounded closure. Writing “B5 T7 unrestricted” or “arbitrary- Q closed” is an over-claim and is forbidden.

3. Why the substantive hypothesis set is empty

Former hypothesis	Disposition (2026-06-12/13)
H-ENDPOINT-THINNESS-ACCEPTED	REMOVED – the per-transfer sunset hardening (SC-SCOPE-SunsetHardened-T6-260612, worst mixed-dressing joint $\times 1.886$) made the endpoint comfortable; thinness retired.
H-NONLATTICE-REMAINDER-EXCLUDED	RECLASSIFIED to definitional scope (route-3, R-037): the beyond-layer bound depends on the competitor only through (T', n) , and Lemma 2 (pure circle-packing) caps $T' \leq 10$ for ALL admissible competitors, lattice or non-lattice. Arbitrary- Q DR-2 only removes the admissibility cap.

Attack-4 audit discharged. The full $(D)(O)(S)$ +budget chain was enumerated (dr2_t030_route3_nonloadbearing.py v1.1.0, assert 6): (D) diagonal isotropy competitor-agnostic; (O) off-diagonal via T' only (Lemma 1 $K_{\text{floor}} \leq T' + \text{Lemma 2 } T' \leq 10$); (S) selection floor competitor-agnostic; K-budget via n only (pattern-generic rectangle). No axis depends on the competitor beyond (T', n) – no hidden geometry dependence.

4. Devil’s-advocate / self-adversarial review: T7-SCOPE is the right tier (CLAUDE.md 6.3)

(α) “Admissibility hides a hypothesis – this is really T6.” ADDRESSED: θ_{min} admissibility is DEFINITIONAL SCOPE (the physical coherence-resolution class, fixed by the microphysics), not a

pending conditional proof, exactly as ‘lattice shells’ / ‘C_full’ is the definitional scope of the B1/B2 head’s own T7-SCOPE. A scoped-T7 with empty substantive conditions is precisely T7-SCOPE, not T6-conditional (which requires a NAMED unproved hypothesis – there is now none).

- (β) “A1-KERNEL-CONV is still a dependency.” VALID and explicit: A1 is the named DEFINITIONAL input (the kernel convention), carried in the scope, identical to the B1/B2 head. It is a definition, not a conditional hypothesis; T7-SCOPE given A1 is the head’s own status.
- (γ) “T-030 is open – how can B5 be T7?” DISMISSED: T-030 (unrestricted arbitrary-Q) is a FRONTIER STRENGTHENING that removes the admissibility cap; it is non-load-bearing for B5’s admissibility-bounded statement (route-3). B5 T7-SCOPE and T-030 OPEN coexist exactly as B1 T7-SCOPE and the open global/all-intensity questions coexist.

Quantitative sanity check (mandatory). The promotion changes NO number: $T' \leq 10$ (Lemma 2), $R_{\text{lead}} = 0.510 < 1$, $K(49) \approx 113 \ll 5972$, sunset worst-dressing joint $\times 1.886 > 1$; the tier action is a classification of the existing evidence under TSv2 (T7-SCOPE = unconditional within the stated definitional scope), with the Attack-4 audit (6/6) confirming the (T', n) -only reduction is complete.

5. Result footer

Result ID:	B5-BeyondLayer-T7Scope-260613 (tier-assignment dossier, route-3 promotion)
Precise statement:	B5-BEYOND-LAYER-BOUND promoted T6 PROVED-CONDITIONAL -> T7-SCOPE on the admissibility-bounded statement: for every admissible competitor (real antipodal, $ Q \leq n_{\text{pack}}$, coherence-resolved $\geq \theta_{\text{min}}$), given A1-KERNEL-CONV, the beyond-layer off-diagonal correction is below the layer margin. Substantive conditional hypotheses: NONE (H-ENDPOINT-THINNESS removed by sunset hardening; H-NONLATTICE reclassified to definitional scope by route-3 R-037, Lemma 2 caps $T' \leq 10$ for all admissible competitors). Attack-4 audit discharged: (D)(0)(S)+budget depend on the competitor only through (T', n) .
Scope:	admissibility-bounded; definitional scope = coherence-resolution class C_adm + A1-KERNEL- CONV. NOT unrestricted/global/arbitrary-Q.
Evidence grade:	T7-SCOPE (unconditional within the definitional scope); EXECUTED (dr2_t030_route3_nonloadbearing.py v1.1.0 6/6; the B5 published bundles).
Reproduction command:	python codes/vacuum/dr2_t030_route3_nonloadbearing.py
Falsifier:	an admissible competitor whose beyond-layer correction exceeds the margin; or a beyond-layer quantity depending on the competitor beyond (T', n) ; or an admissible sum-circle occupancy $> \text{floor}(2\pi/\theta_{\text{min}})$.
Tier before / after:	B5: T6 PROVED-CONDITIONAL -> T7-SCOPE_{admissibility-bounded} (operator route-3 promotion 2026-06-13). NO unrestricted arbitrary-Q claim.
No-overclaim statement:	B5 does NOT claim unrestricted arbitrary-Q DR-2; only admissibility-bounded beyond-layer closure. T-030 stays OPEN as a frontier strengthening; the scope qualifier is

mandatory in every citation.

Next required action: reconcile with the synthesis Sec.4b head finding (B1/B2 T7-SCOPE_{C_full}); B5 is now T7-SCOPE on the same admissible class. T-030 frontier and the unrestricted theorem remain the open research direction.